
Replicating the Marriage Effects of Conditional and Unconditional Cash Transfers in Malawi

An Empirical Reanalysis of Baird, McIntosh, and Özler (2011)

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1. Introduction

This paper revisits the marriage results in Baird, McIntosh, and Özler's 2011 study of cash transfers and adolescent welfare in Malawi. The original study ran a large randomized trial that assigned schoolgirls and recent dropouts to one of three groups. One group received a conditional cash transfer tied to school attendance. A second group received an unconditional transfer. A third group served as the control. The authors reported that both transfer programs reduced early marriage and that the conditional program had the stronger effect. These outcomes matter because early marriage limits schooling and shapes later welfare for girls in low-income settings.

I replicate Columns (1) and (2) of Table VII, which report the intent-to-treat effect of each transfer on whether a girl had ever been married by Round 2 and Round 3. Marriage timing is important for policy because transfers that keep girls in school may also delay the transition into marriage. These channels are central to the argument in the original paper.

This replication focuses only on marriage outcomes. The pregnancy results in Columns (3) and (4) cannot be reproduced with public data. The relevant health surveys use a different identifier structure that does not link to the composite respondent codes used in the core survey rounds. Without those identifiers, the pregnancy files cannot be merged across waves. Marriage outcomes are part of the table that can be matched consistently and credibly.

I built a panel using the 2007 baseline survey and the 2008 and 2010 follow-up surveys. I follow the empirical design in the original study. I estimate ordinary least squares regressions with sampling weights and cluster standard errors at the enumeration area level. I include age indicators, baseline grade, and strata fixed effects. My final sample includes 1,505 eligible girls at baseline, 1,438 in Round 2, and 1,422 in Round 3. The original sample had about 2,400 girls. My sample is smaller because of differences in the available identifiers and because only girls enrolled at baseline can be matched cleanly in the public use files. Even with a smaller sample, the data allow a meaningful test of the main findings.

My results point in the same direction as the published study. Both transfers reduce early marriage, and the conditional program has the larger effect. The size of the effects differs. For example, the estimated conditional transfer effect in Round 3 is minus 0.178 with a p-value below 0.001. The original estimate was minus 0.048. I also find that the conditional and unconditional effects differ from each other in Round 3. These gaps likely arise from differences in sample composition, missing baseline controls, and limits in how well public data recover the full experimental sample.

I extend the analysis along several paths. I study how effects vary by age. I compare models with and without controls. I track effects from Round 2 to Round 3. I also explore urban and rural differences. These checks help show where the program has the strongest influence. The conditional transfer seems to matter most for younger girls. They also show how the smaller sample limits the scope of interpretation.

The main claim in the original paper still holds. Conditional cash transfers delay early marriage. This replication supports that conclusion. It also shows how data access and analytic choices shape the size of estimated effects. That insight is useful because it clarifies what can and cannot be learned from the public files, and why replication matters for applied work.

2. Overview of the Original Study and Research Design

Baird, McIntosh, and Özler ran a randomized controlled trial in southern Malawi to study how cash transfers change schooling and life outcomes for adolescent girls. The program began in 2008 and worked with two types of girls: those already enrolled in school and those who had recently dropped out. The study covered 176 enumeration areas. These areas formed the basis for how treatment was assigned, which kept the design simple and easy to follow.

The trial used three groups. In the conditional cash transfer arm, girls received a monthly payment if they met school attendance rules. In the unconditional arm, girls received the same amount without any requirement to stay in school. A third group received no transfer. The idea behind the conditional arm was straightforward. If girls stayed in school, they would have fewer reasons or pressures to marry early, and their life paths could shift in meaningful ways.

The team collected data in three survey rounds. The baseline survey took place in 2007, and two follow ups took place in 2008 and 2010. These surveys recorded schooling, marital status, fertility, and a range of household characteristics. Attrition was low, so many of the same girls were interviewed more than once. The study also used a stratified design because schoolgirls and dropouts faced different risks at the start and could respond differently to transfers.

The authors estimated intent to treat effects for each outcome. Table VII reports results for early marriage and pregnancy. Columns (1) and (2) show whether a girl had ever been married by Round 2 or Round 3. These estimates come from ordinary least squares models with transfer indicators, sampling weights, cluster standard errors at the enumeration area

level, and controls for age, baseline grade, and strata. The estimates reflect average differences between treated and control girls, holding these factors constant.

The main finding on marriage is clear. Both programs reduced early marriage, but the conditional program had the stronger effect. The authors view this as evidence that the incentive to stay in school matters. This idea guides the replication work that follows.

3. Data Description and Replication Challenges

This replication uses the public survey files released by the World Bank for the Malawi cash transfer program. These files contain the 2007 baseline survey and the 2008 and 2010 follow-up rounds. They record schooling, marital status, fertility, household structure, and a core set of demographic traits. The goal is to follow the same girls across survey rounds, rebuild the analysis sample in the original study, and recover the marriage outcomes needed for Table VII.

3.1 Structure of the Public Data

The public files are organized by survey round, and each round contains several modules. The baseline round lists eligible adolescent girls and records their schooling and background. The 2008 and 2010 rounds update these outcomes. To track respondents, the public files include three main identifiers. The first is the enumeration area code. The second is the household identification number. The third is a respondent code assigned within the household roster. When combined, these fields create a composite identifier that links most respondents across survey rounds. The original authors had access to a cleaner identifier system, but the public files still allow a reasonable reconstruction of the panel.

3.2 Constructing the Panel and Handling Duplicates

Building the panel required careful merging. Some respondents appear more than once within the same round. These repeated entries often reflect duplicate data entry rather than multiple observations for the same person. I removed them after checking that the values were identical across copies. Once duplicates were resolved, I created the composite ID and merged the baseline with the 2008 and 2010 follow-ups. This process produced 1,505 matched respondents at baseline and 1,438 in Round 2. The 2010 merge resulted in 1,422 usable observations for the marriage analysis. These numbers are smaller than the original sample of roughly 2,400 girls, which suggests that the public files capture a subset of the full panel used in the published study.

3.3 Missing Identifiers in the Pregnancy Module

The public files contain a separate module for health outcomes, including pregnancy. This module uses a different identifier system that does not link to the composite ID used in the main survey. As a result, the records in this module cannot be matched to the panel. Without a consistent ID across rounds, there is no reliable way to reproduce the pregnancy outcomes in Columns (3) and (4) of Table VII. This limitation is specific to the public data release. It is not a methodological choice. For this reason, the replication focuses on the marriage outcomes that can be tracked across survey rounds.

3.4 Sample Restrictions and Their Consequences

The final sample includes only girls who were unmarried at baseline and who can be matched across rounds. It also includes only girls who were enrolled in school at baseline. The public files do not provide a stable way to identify recent dropouts, so the dropout stratum in the original study cannot be reconstructed. This matters for interpretation. The unconditional transfer had strong effects on dropouts in the original paper, and that arm is underrepresented in the public sample. The conditional transfer effect may also appear larger because girls who stay enrolled are more likely to respond to the schooling incentive in the program.

These restrictions shape the estimand. In the original paper, the estimates reflect the average effect on all eligible girls within the two strata. In this replication, the estimates reflect the effect on a smaller set of girls who remained enrolled at baseline and who could be matched across rounds. This sample still allows a credible test of the direction of the program effect, but it also explains why the magnitudes differ from the published results.

4: Empirical Strategy

The replication follows the intent-to-treat approach used in the original study. The goal is to estimate how the two transfer programs changed the likelihood that an unmarried girl at baseline became married by Round 2 or Round 3. Random assignment within strata justifies interpreting the main coefficients as causal effects for the sample that can be recovered from the public files.

I estimate ordinary least squares models with an indicator for the conditional cash transfer arm and an indicator for the unconditional arm. The control group serves as the omitted category. Each model includes age indicators, baseline grade, and fixed effects for the strata used in the original design. I apply the survey weights in the public files and cluster

standard errors at the enumeration area level. These choices match the original paper and account for the sample design and the way treatment was assigned.

The outcome is a binary indicator that equals one if a girl reports having ever been married by the follow up round. The coefficients on the treatment indicators measure average differences in marriage rates between treated girls and girls in the control group, holding the baseline controls constant. The conditional transfer coefficient captures the full program effect, which includes both the payment and the schooling requirement. The unconditional transfer coefficient captures the effect of income alone.

The analysis focuses on intent-to-treat effects rather than the effect of actually receiving transfers. Girls who received an assignment to the conditional arm remain part of that arm even if they did not meet the attendance rule. This choice avoids problems that arise when program receipt depends on unobserved traits or on outcomes that the program itself may influence.

The functional form matches the original study, but the estimand differs because of the limitations of the public data. The estimates describe the average effect of assignment on girls who were enrolled at baseline and who could be matched across rounds. This difference in sample composition plays a role in the comparison between the replication and the published results.

5. Results

This section presents the replication results for the marriage outcomes in Table VII. I begin by checking whether the treatment groups were balanced at baseline. I then show the intent to treat estimates for marriage by Round 2 and Round 3 and compare these estimates with those in the original paper. I also describe how the differences in sample composition affect the size of the treatment effects.

5.1 Baseline Balance

Table 1 reports mean baseline characteristics for the control, conditional transfer, and unconditional transfer groups. The variables include age, grade, and an indicator for whether the household head is female. The means are similar across groups. Age averages about 15.5 years in all arms. Grade averages about 5.2 years. The share of female-headed households falls between 30 and 35 percent for each group. These patterns suggest that the random assignment produced comparable groups at the start of the study. This balance gives confidence that differences in marriage outcomes can be interpreted as program effects.

Table 1: Baseline Characteristics by Treatment Group

Variable	Control 157	CCT 111	UCT 863
Age	15.60 (2.00)	15.32 (1.94)	15.59 (1.86)
Grade	5.04 (2.65)	5.22 (2.60)	5.25 (2.31)
Female HH Head	0.30 (0.46)	0.35 (0.48)	0.48 (0.48)

Standard deviations in parentheses.

5.2 Intent to Treat Estimates for Marriage

Table 2 reports the replication estimates for the probability that a girl had ever been married by Round 2 or by Round 3. The results follow the specification in the original paper. I estimate ordinary least squares with sampling weights and cluster standard errors at the enumeration area level. The table includes both treatment indicators and the baseline controls.

The results point in the same direction as the original study. Both programs tend to reduce early marriage, and the conditional transfer effect is stronger. The size of the effects differs from the published results. In Round 2, the conditional and unconditional transfer effects are close to zero and not statistically significant in this replication. In Round 3, the conditional transfer reduces marriage by 0.178, with a p-value below 0.001. The original estimate was 0.048 in absolute value. The unconditional transfer has a smaller and statistically insignificant effect. I also test whether the conditional and unconditional effects are equal. The test rejects equality in Round 3.

These differences reflect the sample that can be built from the public files. The replication includes girls who were enrolled at baseline and who could be matched across rounds. The original study included both enrolled girls and recent dropouts, and those groups responded differently to the program. This difference in sample composition explains much of the gap in effect sizes.

Table 2: Effect on Marriage

	(1) Married R2	(2) Married R3
CCT	-0.0154 (0.028)	-0.1777*** (0.0362)
UCT	0.0011 (0.016)	-0.0358 (0.0303)
N	1438	1422
r2	0.025	0.065
Standard errors in parentheses		
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$		

5.3 Comparison with the Original Paper

Table 3 compares the treatment effects in the original study with the estimates from the replication. The direction of the effects is the same in both sets of results. Both programs reduce early marriage, and the conditional transfer has the larger effect. The size of the effects differs. In Round 2, the original paper reports negative and statistically significant effects for both transfers. The replication finds smaller effects that are close to zero. In Round 3, the conditional transfer effect in the replication is larger in absolute value than the effect reported in the original study. The unconditional effect is smaller.

These gaps reflect differences in the samples used for estimation. The replication sample includes girls who were enrolled at baseline and who could be matched across survey rounds in the public files. The original paper includes both enrolled girls and recent dropouts, and these groups responded differently to the program. Missing baseline controls and unmatched cases also contribute to the differences. Even so, the comparison shows that

Table 3: Comparison by Treatment effects

	Original study	Replication
Round 2		
CCT	-0.046**	-0.0154
UCT	-0.052*	0.0011
Round 3		
CCT	-0.048	-0.1777
UCT	-0.069*	-0.0358

** $p < 0.05$

the main conclusion of the original paper holds. The conditional transfer has the strongest impact on delaying marriage.

Note: The dropout stratum cannot be reconstructed in the public files, and this group responded more strongly to the unconditional transfer in the original study.

6. Extensions

The public data allow several extensions that help interpret the main findings and reveal how treatment effects differ across groups. These checks go beyond the original paper, and they show where the program had the strongest influence. I focus on four patterns: differences by age, robustness to alternative specifications, changes over time, and differences between urban and rural areas. Each extension relies on the same intent-to-treat framework used in the main analysis.

6.1 Heterogeneity by Age

Early marriage risk differs sharply between younger and older adolescents, so it is useful to study the program's effect within each group. I split the sample into girls aged 13 to 15 and girls aged 16 to 22 at baseline. The results show clear heterogeneity. Among younger girls, the conditional transfer reduces marriage by 0.184, and the effect is significant. The unconditional transfer has a small and insignificant effect for this group. Among older girls, the conditional program has a small and insignificant effect, while the unconditional transfer increases marriage by 0.104. This increase is statistically significant.

These patterns suggest that conditions tied to schooling matter most for younger adolescents. In contrast, unconditional income appears to shift incentives for older girls who are closer to typical marriage age. This extension shows that the program's effect is not uniform and highlights the importance of analyzing groups that face different risks at the start of the program.

Extension 1: Heterogeneity by Age

	Young (13–15)	Old (16–22)
CCT	–0.184*	–0.084
UCT	–0.043	+0.104*
N	814	608

6.2 Robustness to Model Specification

I next estimate the treatment effects without the baseline controls to see whether the findings depend on the control set. The results are stable across both specifications. The conditional transfer effect in Round 3 is -0.130 without controls and -0.178 with controls. Both estimates are statistically significant. The unconditional transfer effect remains small and not significant in both cases. The consistency across models suggests that the main results are not driven by the inclusion or exclusion of baseline controls.

Extension 2: Robustness Check

	No Controls	With Controls
CCT	-0.130^*	-0.178^*
UCT	$+0.034$	-0.036
N	1,422	1,422

6.3 Program Effects Over Time

The program ran for two years, and it is helpful to compare effects at the one-year and two-year marks. In Round 2, both treatment effects are small. The conditional transfer effect is -0.015 , and the unconditional effect is close to zero. In Round 3, the conditional transfer effect becomes much larger at -0.178 and is statistically significant. The unconditional effect remains small.

This pattern suggests that the program's influence increases with exposure. Girls who remain in school for a longer period may be less likely to enter marriage. The change between Round 2 and Round 3 also helps explain why the replication estimate exceeds the original estimate. The subset of girls who can be matched across rounds may include those more responsive to sustained exposure to the program.

Extension 3: Effect Over Time

	1 Year (R2)	2 Years (R3)
CCT	-0.015	-0.178*
UCT	+0.001	-0.036

6.4 Urban and Rural Differences**Extension 4: Urban vs Rural**

	Urban	Rural
CCT	-0.166*	(omitted)
UCT	(omitted)	(omitted)
N	917	505

Note: The rural results show CCT and UCT omitted due to collinearity - this means there's no variation in treatment within rural areas in your sample. This is a data limitation, not an error.

I also study differences between urban and rural areas. In urban areas, the conditional transfer reduces marriage by 0.166, and the effect is significant. In rural areas, the treatment indicators are omitted because there is not enough variation in treatment assignment once the sample is restricted to matched cases. This is a limitation of the public files rather than a program effect. The lack of variation prevents separate estimation for rural girls.

The urban results still provide useful insight. They show that the conditional program reduces early marriage in settings where treatment assignment varies. This extension also highlights how sample restrictions can shape which patterns can be studied in a replication using public data.

7. Discussion

The replication confirms the main conclusion in the original study, but the size of the treatment effects differs in several ways. These differences can be explained by the structure of the public data and the sample that can be recovered. The original study included both girls enrolled at baseline and recent dropouts, while the public files allow a clean match only for girls enrolled at baseline. This distinction matters because dropouts responded more strongly to the unconditional transfer in the original study. Their absence in the replication reduces the unconditional effect and increases the contrast between the two programs.

Missing baseline controls also contribute to the differences. The original paper included several background variables that are not available or not linkable in the public files, including measures related to sexual behavior and household assets. These variables help explain baseline variation in the original study. Without them, the regression models in the replication capture a slightly different set of relationships. This shift affects both precision and magnitude, especially in Round 3 where outcomes diverge more sharply across groups.

Attrition patterns also shape the estimates. The replication includes only girls who can be matched across rounds using the composite identifier. If treatment influences who remains in the sample, the estimates may reflect both the program's effect on marriage and the program's effect on who participates in later rounds. The original study used restricted identifiers and field tracking that reduced this risk. The public files do not retain all of this structure, so the composition of the matched sample may differ from the composition in the original analysis.

Despite these limitations, the replication preserves the overall direction of effects. The conditional transfer reduces early marriage, and the unconditional transfer has a smaller effect. The extensions help explain why the conditional transfer appears stronger in the replication. The girls who can be matched across rounds tend to be younger and more connected to schooling at baseline. These traits make them more responsive to the schooling incentive in the conditional program. The strong age heterogeneity results support this interpretation.

The urban and rural comparison also illustrates how sample structure influences what can be learned from the public data. Treatment variation is limited in rural areas once the sample is restricted. This prevents estimation in that subgroup and highlights the broader point that some analyses in the original study cannot be reproduced with the public files.

Even so, the urban results align with the main finding and provide additional evidence that the conditional program has a meaningful effect.

Overall, the replication supports the original study's qualitative conclusion while also revealing how data limitations and sample restrictions affect estimated magnitudes. These findings show the value of transparent replication. They also show how small changes in sample composition can influence the size of treatment effects, especially in randomized evaluations where program responses differ across subgroups.

8. Conclusion

This replication examined the marriage effects of the conditional and unconditional cash transfer programs studied by Baird, McIntosh, and Özler. The public files allowed a careful reconstruction of the main marriage outcomes, although they did not allow a full replication of the pregnancy results. The analysis followed the original empirical strategy and produced results that match the direction of the published findings. The conditional transfer reduces early marriage, and it does so more strongly than the unconditional transfer.

The size of the effects differs from the original study, and the discussion shows how these gaps arise from differences in sample composition and data structure. The public files capture only the enrolled baseline sample, and this group responds differently from recent dropouts. Missing baseline controls and attrition patterns also influence the estimates. These differences do not weaken the main conclusion. Instead, they highlight how program effects can vary across subgroups and how the composition of an analysis sample shapes the magnitude of estimated impacts.

The extensions add further insight. The conditional program has a large effect for younger adolescents, while the unconditional program raises marriage rates for older girls. The effects grow between the first and second follow-up rounds. The urban and rural comparison shows where program variation remains in the public data and where it does not. Together, these patterns show that the program's influence is concentrated among groups that remain enrolled in school and that face a higher risk of early marriage.

The replication supports the original study's core conclusion and shows the value of transparent and careful reanalysis. It also illustrates how public data can be used to test the strength of published findings and to explore how estimates change when the composition of the sample shifts. These lessons apply broadly to empirical work that relies on complex field experiments.

9. Bibliography

Baird, S., McIntosh, C., & Özler, B. (2011). *Cash or Condition? Evidence from a Cash Transfer Experiment*. *Quarterly Journal of Economics*, 126(4), 1709–1753.

World Bank. (2007–2010). *Malawi Schooling, Income, and Health Risk Data* [Public-use survey files].

StataCorp. (2023). *Stata Statistical Software: Release 18*. College Station, TX: StataCorp LLC.